Postprandial Glucose Response Prediction Using CGM and Nutritional Data

A Comparative Study of Machine Learning Approaches

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Introduction

Motivation & Research Questions

Problem Domain & Clinical Motivation

What is Postprandial Glucose Response?

- Blood glucose rises after a meal, peaks ~1 h, returns to baseline in 2–3 h
- Thresholds (2-h): <140 mg/dL **normal**, 140–199 mg/dL **pre-diabetes**, ≥200 **T2D**

What shapes the response?

- Not carbohydrates alone → many factors interact
- Carbs, fiber, fat, protein - plus activity, insulin sensitivity, microbiome

Why prediction is hard?

- Zeevi et al. (2015) [1] - the same meal gives very different responses across people
- Prediction must be personalized

[1] - David Zeevi, Tal Korem, Niv Zmora, and 1 others. 2015. Personalized nutrition by prediction of glycemic responses.

Related Work

Data availability

- Most public CGM datasets focus on type-1 diabetes
- Healthy + T2D data with detailed meal logs remains rare

Classical ML approaches

- Zeevi (2015) [1] & Mendes-Soares (2019) [2] : gradient boosting predicts PPGR across cohorts
- RF, XGBoost, LightGBM, SVR widely used for non-linear feature interactions

The research gap

- No consistent comparison across model families
- Benefit of meal-type structure not systematically tested

[1] - David Zeevi, Tal Korem, Niv Zmora, and 1 others. 2015. Personalized nutrition by prediction of glycemic responses.
[2] - Helena Mendes-Soares and 1 others. 2019. Assessment of a personalized approach to predicting postprandial glycemic responses to food among individuals without diabetes.

Research Question and Contributions



Research question

How do classical ML, mixed-effects, and foundation models compare for PPGR prediction, and does stratifying by meal type help?



Comparison & evaluation

Compare three model families (linear mixed-effects, XGBoost, TabPFN) under LOSO, on two targets (AUC, iAUC) and evaluate meal-type stratification



Interpretation

Cross-model SHAP feature importance and prediction behaviour across cohorts

02

Data Preprocessing

CGMacros Dataset



Study Cohort

- 45 adults, 10 days (Sansum Diabetes Research Institute, Santa Barbara CA)
- 3 metabolic groups: Healthy (15), Pre-Diabetic (16), T2D (14)
- Two blinded CGM devices + Fitbit Sense + meal logs with before/after photos



Controlled Meal Protocol

- **Breakfast**: standardized protein shake (fixed macronutrient composition)
- **Lunch**: from a fixed restaurant chain (documented nutritional content)
- **Dinner**: freely chosen, self-reported (least reliable)
- **Snacks** allowed in between meals



Multimodal Data

- **Fasting blood panel**: HbA1c, glucose, insulin, lipids, BMI
- **Gut microbiome**: Viome sequencing + 22 composite health scores

Data Quality and Filtering

CGM Device Selection

- Dexcom Device excluded: systematic offset up to 40 mg/dL + noisier 5-min trace
- All experiments use Freestyle Libre only (complete data recordings for all subjects)

Filtering steps

- **1 subject removed**: Fitbit not worn (MNAR)
- **Snacks removed**: sparse and nutritionally heterogeneous
- **Outlier meals removed**: zero calories, invalid macros values, glucose < 40 mg/dL

Final Dataset

- 1170 meals, 44 subjects
- Healthy: 15, Pre-Diabetic: 15, T2D: 14

Feature Engineering

Prediction Targets

- **AUC**: total glucose exposure over 2h after a meal (reflects baseline level and the rise)
- **iAUC**: how much glucose rose above the pre-meal level (isolates the postprandial spike)

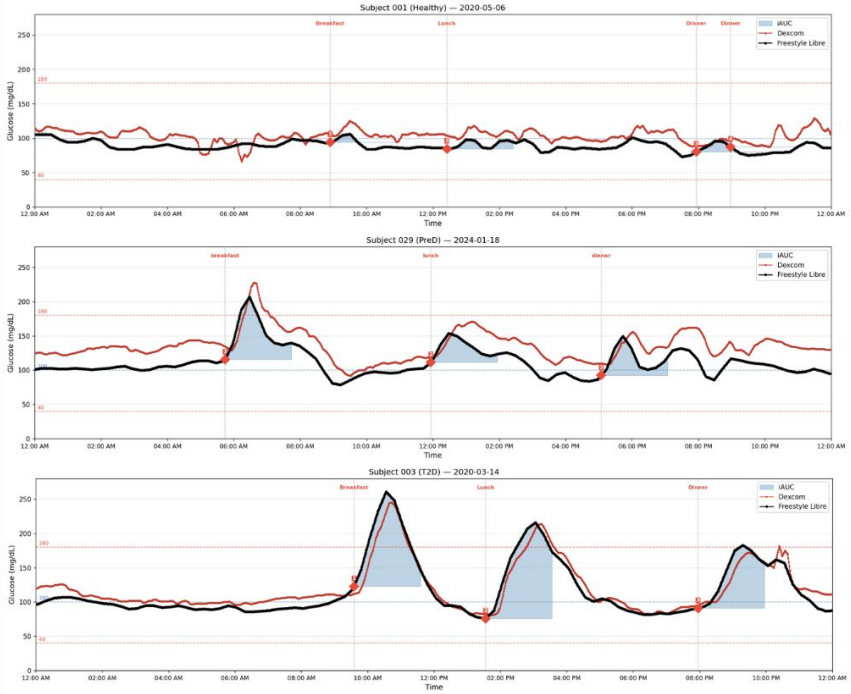
Derived Features

- HOMA-IR
- Pre Meal Activity (1h)
- Post Meal Activity (2h) (calorie expenditure)

Final Feature Set

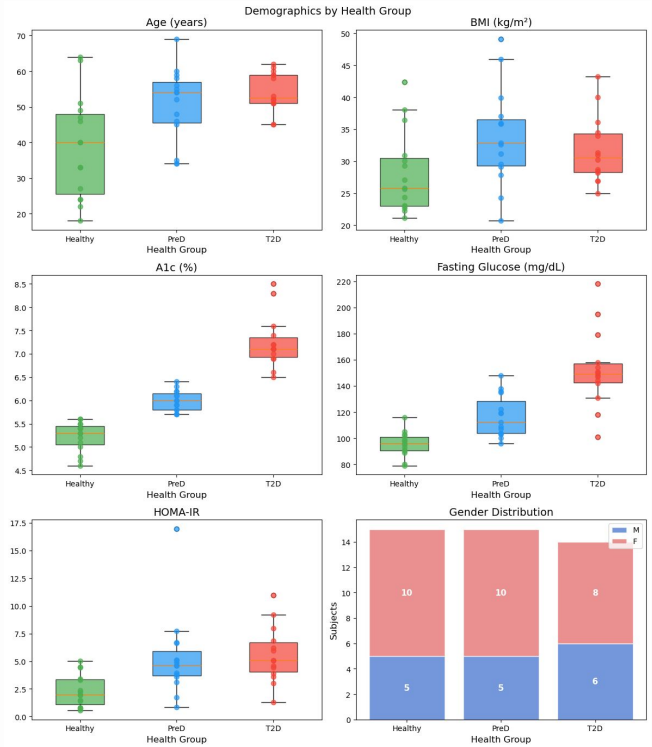
Category	Feature
Meal composition	Carbs, Protein, Fat, Fiber (kcal) Consumed Calories (kcal)
CGM	Baseline Libre glucose (mg/dL)
Clinical	Age, Gender, BMI HbA1c, Fasting glucose, Insulin HOMA-IR, Triglycerides Cholesterol, HDL, Non-HDL, LDL, VLDL, Cho/HDL ratio
Activity	Activity_pre_sum (kcal, 1hr) Activity_post_sum(kcal, 2hr)

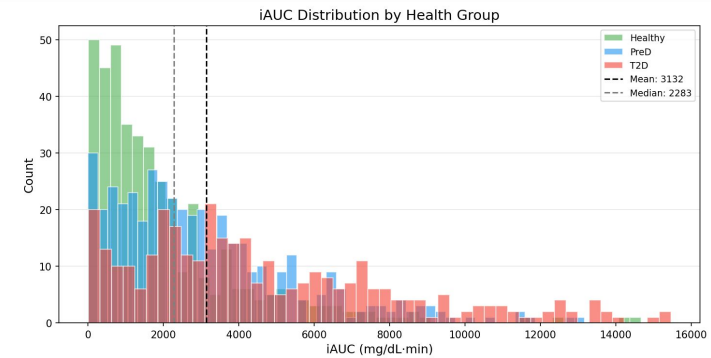
Exploratory Data Analysis

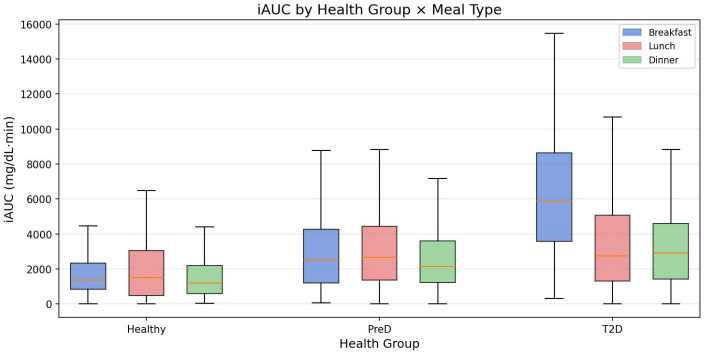


- **24-hour CGM recordings** for one representative participant from each metabolic group (Healthy, Pre-diabetic, T2D).
- The Freestyle Libre (black) and Dexcom (red) traces are shown alongside meal onset markers.
- Shaded regions indicate the incremental AUC
- (iAUC) computed above the pre-meal Libre baseline for each meal.
- Reference lines mark the sensor floor (40 mg/dL), normal fasting glucose (100 mg/dL), and the hyperglycemia threshold (180 mg/dL)

Exploratory Data Analysis







03

Methods

Experiments & Evaluation

Experimental Design & Evaluation

Experimental Design

- **Leave-one-subject-out (LOSO)**: all meals from one subject held out, trained on the other 43
- **StandardScaler** fit on the training fold only: no normalization leakage, metrics aggregated across folds

Prediction Targets

- **AUC**: total area under the curve, includes baseline
- **iAUC**: incremental area above pre-meal baseline

Evaluation Metrics

- **Evaluation Metrics**: Pearson r (primary), RMSE, MAE, R^2
- **Explainability**: SHAP (TreeExplainer for XGBoost, Permutation for TabPFN) & LMM coefficients

Models Evaluated

Linear Models: OLS & Mixed-Effects

- OLS baseline
- LMM adds a random intercept per subject for individual differences
- Fitted with statsmodels

Variants:

- **LMM-base**: + random intercept / subject
- **LMM-interaction**: meal × health group
- **LMM-stratified**: separate B/L/D models

XGBoost

- 80 trees, lr 0.2, depth-1 stumps, L1 regularization (same config as CGMacros)
- Four configurations:
  - Original
  - Extended (+Activity)
  - Stratified (Breakfast/Lunch/Dinner)
  - General (all meals)

TabPFN

- Transformer-based foundation model for tabular data
- In-context learning: no task-specific training or hyperparameter tuning
- Built-in preprocessing and missing-value handling
- Evaluated with the same General and Stratified configurations; designed for small datasets

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Results

I. Linear Models Results

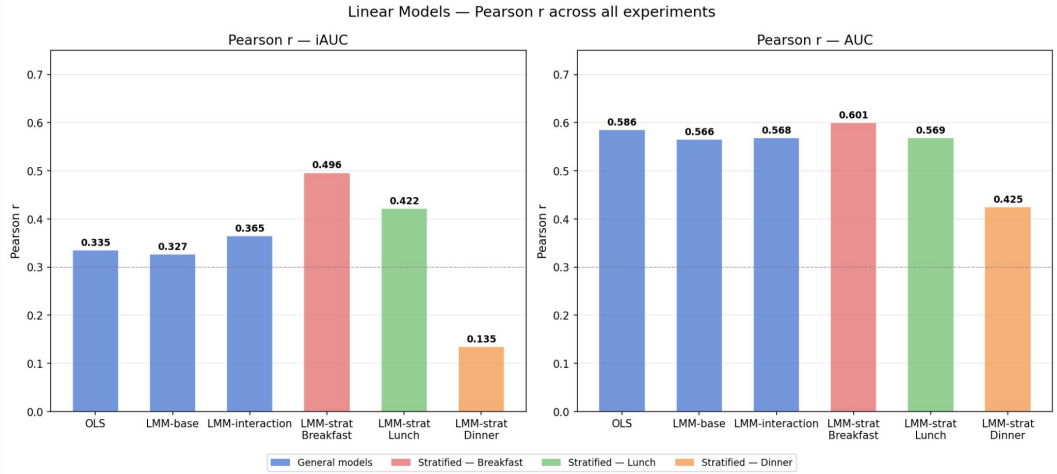
Key Takeaways

- iAUC: all general models weak (r ≈ 0.33–0.37) (linear models struggle with the reactive spike)
- AUC: stronger across the board - OLS best at r = 0.586
- Stratification helps: Breakfast LMM reaches r = 0.496 (iAUC), 0.601 (AUC)
- Dinner prediction fails - r = 0.135, R^2 = −0.321

Why did LMMs underperform?

- Random intercept cannot be estimated for a held-out subject
- Model falls back to fixed effects only - but miscalibrated
- LOSO and LMMs are a known mismatch

Model	r	RMSE	MAE	R^2
iAUC				
OLS	0.335	2859	2055	0.016
LMM-base	0.327	2915	2091	-0.023
LMM-interaction	0.365	2884	2056	-0.001
LMM-strat Breakfast	**0.496**	**3181**	**2215**	**0.143**
LMM-strat Lunch	0.422	2548	1884	0.088
LMM-strat Dinner	0.135	2518	1782	-0.321
AUC				
OLS	0.586	4500	2577	0.192
LMM-base	0.566	4565	2655	0.169
LMM-interaction	0.568	4561	2638	0.170
LMM-strat Breakfast	**0.601**	**5328**	**2830**	**0.223**
LMM-strat Lunch	0.569	3924	2407	0.126
LMM-strat Dinner	0.425	3962	2328	-0.127



I. Linear Models Explainability

Feature	β	Interpretation
Meal type (Lunch)	−1083	Breakfast produces highest spikes
Meal type (Dinner)	−1357	Breakfast produces highest spikes
HbA1c	+1984	Worse metabolic status → larger spike
Baseline glucose	−28.2	Higher pre-meal GL → less room to rise
Post-meal activity	−2.85	Activity attenuates the spike
Protein	−4.4	Known glucose-attenuating effect

LMM Full Dataset Inference

- Fitted on all 44 subjects to examine significant predictors
- 5 significant features ($p < 0.05$)

II. XGBoost Results

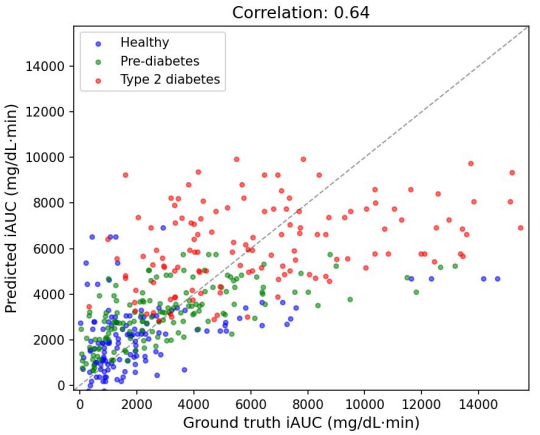
Activity Experiment

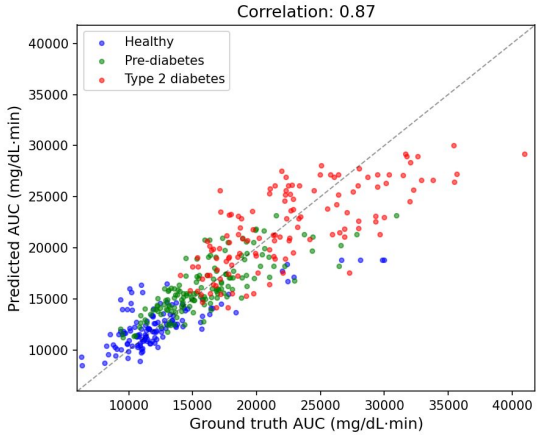
Activity features slightly improve iAUC prediction
AUC remains essentially unchanged
Original configuration reproduces CGMacros benchmark

- r = 0.642 vs 0.645

T2D subjects show the largest errors → extreme spikes hardest to predict

Config	iAUCr	AUCr
Original(19 features)	0.642	0.873
Extended(+activity)	0.652	0.875
Das et al.(2025)	0.645	0.885





II. XGBoost Results

Meal Stratification Experiments

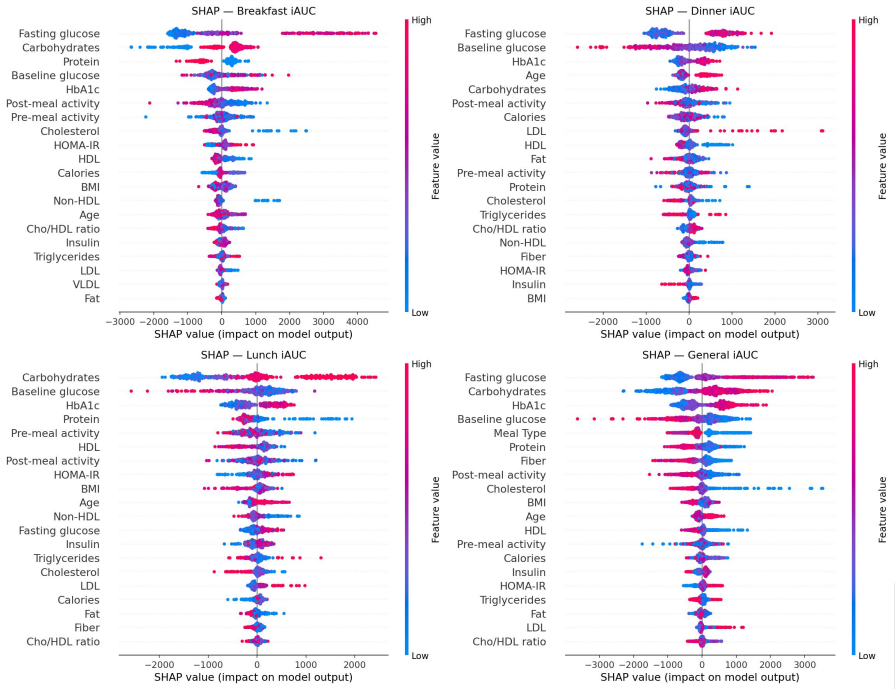
Configuration	r	RMSE	MAE	R^2
iAUC				
Breakfast ($n = 405$)	0.655	2597	1877	0.429
Lunch ($n = 369$)	0.488	2355	1808	0.220
Dinner ($n = 396$)	0.257	2171	1567	0.018
General ($n = 1170$)	0.469	2550	1878	0.217
AUC				
Breakfast ($n = 405$)	0.877	2943	2158	0.763
Lunch ($n = 369$)	0.794	2564	1945	0.627
Dinner ($n = 396$)	0.729	2557	1866	0.526
General ($n = 1170$)	0.829	2822	2072	0.682

- AUC is predictable regardless of meal type even dinner reaches r = 0.729
- iAUC tells a different story: performance degrades sharply from breakfast → lunch → dinner
- **Dinner iAUC (r = 0.257) is barely above chance**: self-reported meals + day-long variability accumulation
- General model is a reasonable compromise for AUC, but not for iAUC
- **Meal structure matters more for the reactive spike** (iAUC) than for total glucose burden (AUC)

II. XGBoost Explainability: iAUC

What drives the reactive spike?

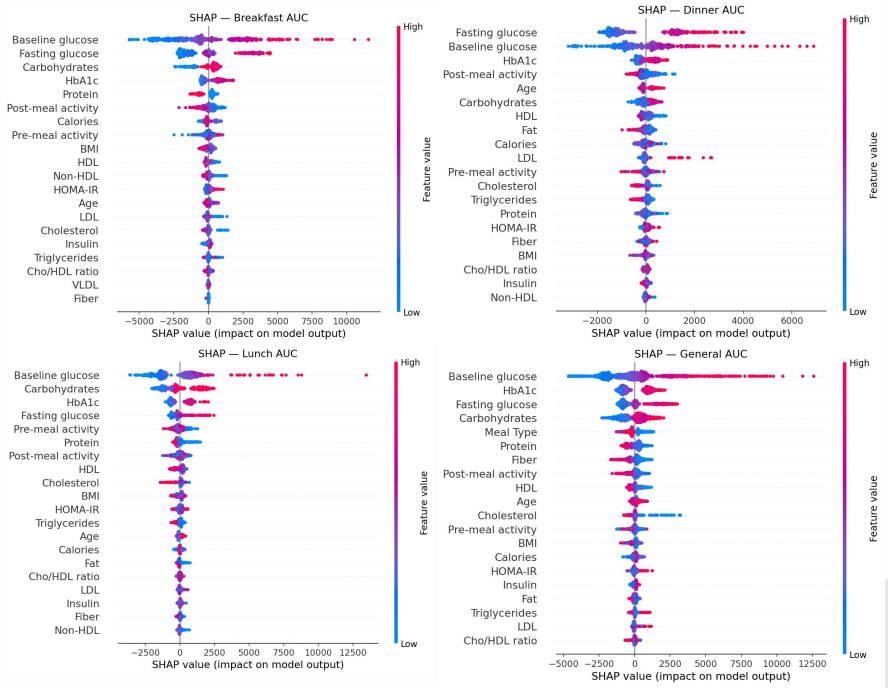
- Top features shift by meal type - no single universal predictor
- **Breakfast**: fasting glucose dominates - chronic metabolic state matters most
- **Lunch**: carbohydrates take the top position - meal composition becomes the main driver
- **Dinner**: fasting + baseline glucose + HbA1c lead → carbs rank lower, model relies more on metabolic state because meal **data is less reliable**
- **Activity features** (pre + post meal) appear in top 10 across all models - validates our feature engineering choice



II. XGBoost Explainability: AUC

What drives total glucose exposure?

- **Baseline glucose** is the single dominant feature across all meal types - if you start high, you end high
- **HbA1c and fasting glucose** follow consistently - chronic glycaemic control is the main signal
- **Carbohydrates** rank lower than for iAUC - nutritional content matters more for the spike than for total burden
- In the **General model**, **meal type itself is an important feature** - the model learns that breakfast, lunch, and dinner produce systematically different glucose profiles



III. TabPFN Results

Key Takeaways

- Same meal-reliability trend: breakfast best, dinner weakest
- AUC consistently strong across all meals, even dinner (r = 0.786)
- Adding meal type to general model improves iAUC (0.571 → 0.593)
- TabPFN slightly edges XGBoost on breakfast, with no tuning required

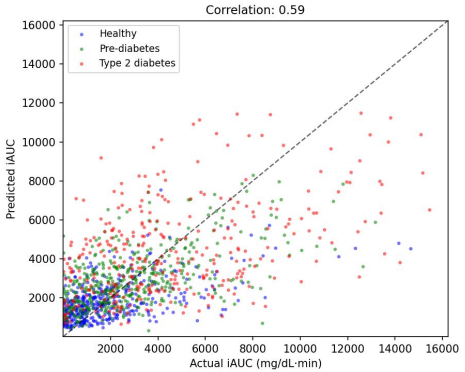
Plot iAUC:

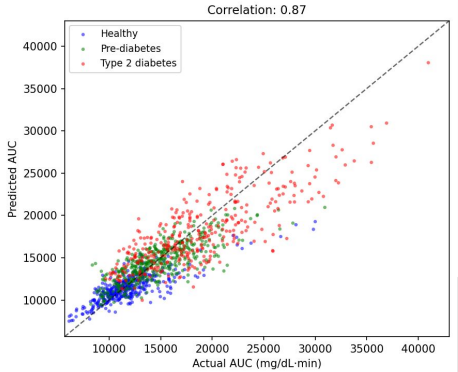
- Model underpredicts large spikes - T2D subjects show largest errors
- Healthy subjects cluster near origin, easier to predict

Plot AUC:

- Points tight along diagonal across all metabolic groups - AUC is the easier target

Configuration	r	RMSE	MAE	R^2
iAUC				
Breakfast	0.661	2593	1871	0.430
Lunch	0.529	2301	1724	0.256
Dinner	0.336	2124	1545	0.060
General	0.571	2389	1727	0.313
General+MT	0.593	2341	1698	0.340
AUC				
Breakfast	0.892	2795	2005	0.786
Lunch	0.796	2544	1915	0.633
Dinner	0.786	2301	1698	0.616
General	0.860	2634	1904	0.723
General+MT	0.867	2563	1858	0.738





III. TabPFN Explainability

Key Findings

For iAUC (General)

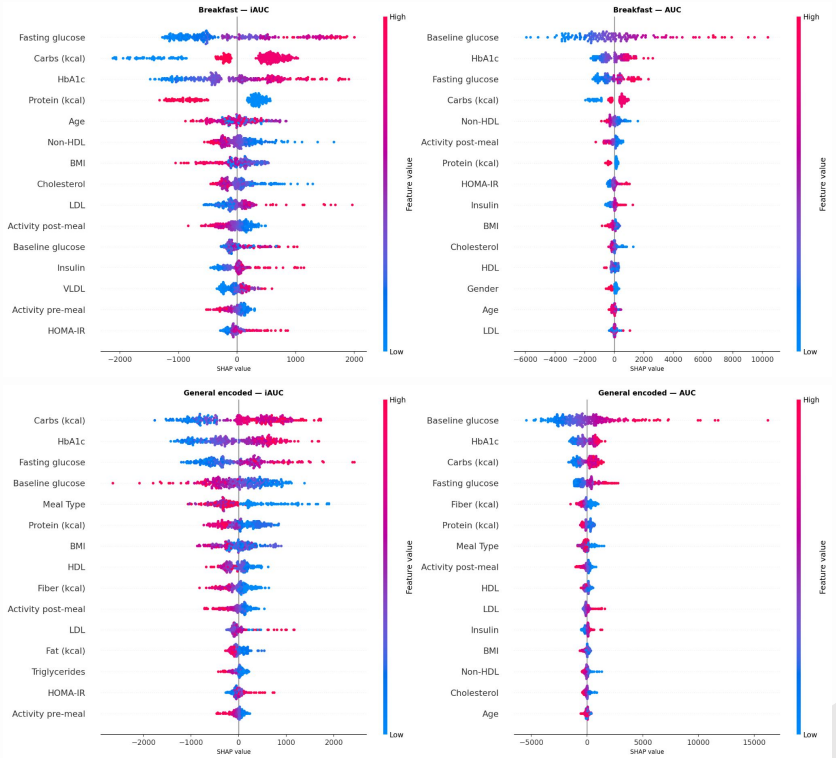
- Carbohydrates are the strongest predictor
- HbA1c and fasting glucose reflect metabolic status
- Post-meal activity reduces predicted glucose spikes
- Meal type contributes additional predictive information

For iAUC (Breakfast)

- Fasting glucose becomes the top predictor
- Individual metabolic state matters more when meals are standardized

For AUC

- Baseline glucose is the dominant predictor
- Feature importance is consistent with XGBoost



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Discussion & Conclusion

Key Takeaways



Findings

Meal context governs accuracy

Across all families, breakfast (standardized) predicts best; dinner (free-form) worst - a dataset-structure effect

Activity helps modestly

Pre/post-meal activity adds a small but real improvement, mainly for iAUC

TabPFN ≈ XGBoost

A foundation model matches tuned boosting on a small biomedical table - without model-specific tuning

Drivers are consistent

Carbs, HbA1c, fasting & baseline glucose recur as top predictors across models

Limitations & Future Directions

Limitations

Small cohort (44 subjects): performance estimates are less stable

Missing biological markers: insulin dynamics and microbiome not yet used

Meal categories differ in data consistency: dinner self-reported, controlled breakfast

Single cohort: no external validation of generalization

Future Directions

CGM time-series foundation models: learn directly from glucose trajectories rather than AUC/iAUC summaries

Integrate biological signals: incorporate gut microbiome features to improve personalized glucose-response prediction

Conclusion

Predicting PPGR is feasible

- PPGR can be predicted from CGM, nutritional, clinical, and activity data
- Performance depends on both the meal type and the prediction target

Model comparison

- Linear mixed-effects models provide an interpretable baseline
- Nonlinear models achieve higher predictive performance
- TabPFN performs comparably to XGBoost on small biomedical datasets

Future improvements

- Better prediction requires capturing individual variability
- Larger cohorts, richer biological information, and CGM time-series data may improve generalizability

Thank you!